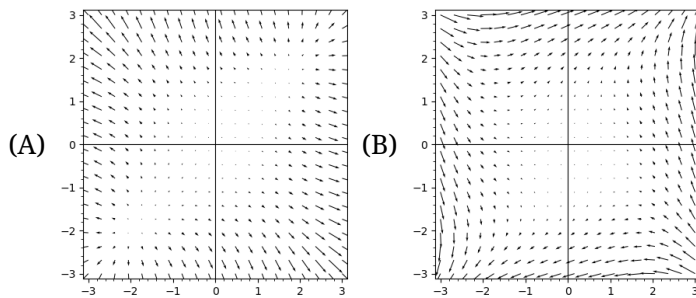


Complex Variable - Midterm Mock Exam

考試時間 Test time 10:30:00 am to 12:00:00 pm; 1.5 hours. 以教室投影機時鐘為準 Using the projected clock.
用空白紙張作答 Answer on blank paper. 每個問號一分 One point per question mark. 全對才給分 No partial credit.
可以帶 991 You can use fx-991 or anything weaker. 舉手問問題 Raise your hand to ask questions.

[1] A vector is ∇ -free if $P_x + Q_y = 0$. Which of the following picture is this-free?



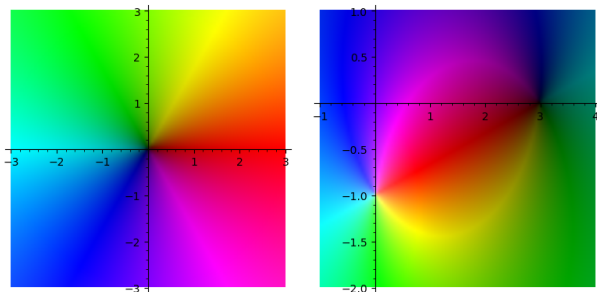
[2] A vector is ∇ -free if $P_y = Q_x$. Which of the picture above is this-free?

[3] Select all that apply. For $u: \mathbb{R}^2 \rightarrow \mathbb{R}$ a harmonic function, The vector field (u_x, u_y) is (C) divergence-free (D) curl-free (E) path dependent (F) diffeomorphic?

[4] Consider using the anti-derivative of $P - Qi$ to solve for fluid dynamics problems. Which are correct? (G) fluid flows faster when the Ψ contours are closer. (H) fluid flows faster when the Φ contours are closer. (I) fluid flows faster when the Ψ contours are farther. (J) fluid flows faster when the Φ contours are farther.

[5] What condition is not needed when proving that a subset of $\mathbb{C}$ is $\mathbb{C}$ itself? (K) nonempty (L) open (M) close (N) compact

[6] The lower left picture is the domain coloring plot of $h(z) = z$. The lower picture is $?(? - z)/(z + ?)$. (All question marks are Gaussian integers.)

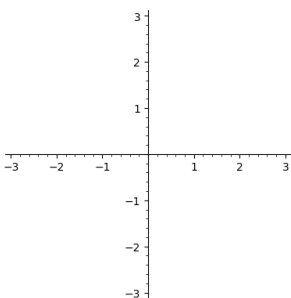


[7] Decompose the Möbius map $d(z - a)/(z - b)$ into to a composition of $+c$, $c\cdot$, and $^{-1}$.

$$\mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C}$$

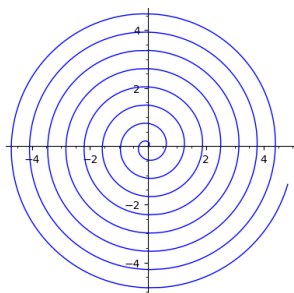
[8] Consider the Möbius map $g(z) = 6/(7 - z)$. Let G be the composition of 10 g 's, then $1/G'(1) = 60466???$.

[9] The conformal map $z \mapsto z + 1/z$ maps what to the upper half plane? Draw on the canvas.



[10] $\exp(2000 + i) = ????701060846160 \cdot 10^{868} + ????590052030137 \cdot 10^{868}i$.

[11] Define a complex $\ln$ function such that $\ln(1) = 0$ and is continuous at points not on the spiral. What is $\ln(3 + 3i)$???



[12] Let f be an arbitrary holomorphic function. The integral $\int_0^\infty \frac{f(x^3)}{1+x^3} dx$ can be expressed using f as ??? (Do not use the integral symbol in your answer.)

[13] Consider an integral of the form $\int_{-1}^2 f(x) dx$. If f is meromorphic and has a pole at $x = 0$, then it probably does not converge. Its (Cauchy) principal value is defined as $\lim_{\delta \rightarrow 0} \int_{-1}^{-\delta} f(x) dx + \int_{\delta}^2 f(x) dx$. For $h(x) = \sin(x)/x^4$, the principal value is 0.???259575107558.

[14] The Schwarz–Christoffel formula $f(\zeta) = \int_0^\zeta (z+1)^{-?}(z+2)^{-?} dz$ maps the upper half plane to an equilateral triangle. The number ? is mapped to the lower left corner. The number ? is mapped to the lower right corner. The number ? is mapped to the top corner. The side length is ?.2999162508563.

[15] List three (???) bijective conformal maps from the upper half plane to the triangle in the previous problem, and they must be different from the f above. If there are not enough functions, write down NOMORE.

[16] For a harmonic function $u: \mathbf{C} \rightarrow \mathbf{R}$, there exists another harmonic function $v: \mathbf{C} \rightarrow \mathbf{R}$ such that $u + iv$ is holomorphic. Let $u(x, y) = -(y^2 - x^2 - 1)/(4y^2x^2 + (y^2 - x^2 - 1)^2)$ and $v(0, 0) = 0$. Then $v(1, 1.5) = -0.???034482758621$.

[17] Consider the Dirichlet problem on a circle. That is, u is harmonic inside the circle, continuously extending to the boundary, and $u(e^{i\theta}) = \beta(\theta)$ for the boundary condition $\beta(\theta) = \cos(\theta)^{72}$. Then $u(0 + 0i) = 0.???770122224943$.

[18] If f and g are holomorphic functions on the entire $\mathbf{C}$ and $f(1/n) = g(1/n)$ for all positive integer n , why is $f = g$ on the entire $\mathbf{C}$? (Tell me the name of the theorem used.) If f and g are holomorphic on the punctured plane $\mathbf{C} \setminus \{0\}$, does $f = g$ still hold or you can find a counterexample???